

Systems of Linear Equations



Setting up the augmented matrix

- 1 Write the augmented matrix for the system $2x + y - z = 3$, $x - 3y + 2z = -1$, $4x + y + z = 5$ (give it as its three rows).
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Elementary row operations

- 2 List the three types of elementary row operations.
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Solving 2x2 systems by elimination

- 3 Solve $2x + 3y = 7$, $x - y = 1$.
 - 4 Solve $3x - 2y = 4$, $5x + y = 11$.
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Solving 3x3 systems by Gaussian elimination

- 5 Solve $x + y + z = 6$, $2y + 5z = -4$, $2x + 5y - z = 27$.
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Why row operations preserve the solution set

- 6 Explain briefly why multiplying a row by a nonzero constant C doesn't change a system's solution set.
- 7 Why is the restriction $c \neq 0$ needed for this row operation?